based company, is investing in 3D printing for component manufacturing, in order to create better, faster and more cost effective parts for its products.

In the automotive industry Ford Motors is planning to create a system whereby their customers would be able to print their own replacement parts. Traditionally, replacement parts need to be shipped from the factory to mechanics or service centres and can lead to days of work. But Ford wants to find a way so that customers can download the designs of the parts locally and have it printed at home or at the service centre within hours.

Till such a time as that is possible Ford is using 3D printing to run prototyping on the components test vehicles and has been doing so since the late 198Os. Using large scale industrial grade 3D printers, Ford saves countless amount of man hours when testing and designing components for its cars, engines and other automotive systems. On average, Ford is able to save one month of production time on certain lines of engines, which include a complex array of features such as ports, passages and valves with no loss in quality. The ability to quickly create and test these prototypes allows them to work towards not only enhance fuel efficiency designs but also experiment with a variety of designs in one rotation. Traditional manufacturing based prototyping would take up to four or five months per component.

Mattel Toys uses dozens of 3D printers to make prototypes of its toys using clay and wax, before the final model is rendered in plastic. Nearly every product made by Mattel incorporates 3D printed components including the popular Barbie and Hot Wheel toys. Other end user companies like shoe manufacturers Nike and New Balance are able to design sports shoes which are built according to the specific feet scans of athletes allowing for greater comfort, safety and speed. In some respects, this inclusion of 3D printing in apparel and shoes is hopeful for a world without sweatshops.

Consumer use and Customised Manufacturing

And even as larger industries are considering how the benefits of 3D printing can be incorporated further into their business models, it is at the level of consumer business' where the highest levels of engagement is seen. The idea of mass customization allows for customers to directly influence the style, look, shape and features of their products. From jewellery to sports good, every customer can demand a distinct identity for their personal products. Small business' in the U.S. are already tapping into this desire with the features of 3D printing. Traditionally, customization has always been the last step of any products creation chain, such as a photo print